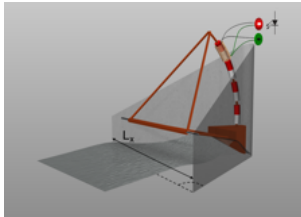


On designing and modelling contraction and sheet wave-energy devices

[Onno Bokhove](#) with Grieve, Lee, Naar, Thompson

Seminar at Department of Physics, University of Patras, Greece

School of Mathematics, Leeds Institute for Fluid Dynamics, UK



Delft-Toronto-Bristol-Twente-Leeds

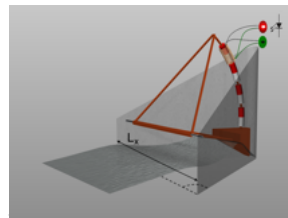
- ▶ Delft: Molecular dynamics & statistical physics (OB et al. J. Stat. Phys., 1994), incl. Lynden-Bell's work.
- ▶ Toronto: Theoretical dynamic meteorology, chaotic Lorenz-1986 system, symplectic integrators.
- ▶ Bristol: Shallow-water simulations and magma dynamics (finite-difference hyperbolic systems).
- ▶ Twente: Water waves, variational/Hamiltonian dynamics and discontinuous Galerkin FEM.
- ▶ Twente/Leeds: Outreach experiments: bore-soliton-splash demonstrator <https://www.youtube.com/watch?v=YSXsXNX4zW0> to Wetropolis flood <https://www.youtube.com/watch?v=yUjYfg2SfY0>.
- ▶ Leeds: DA, flood-mitigation cost-effectiveness tool (after 2015 Yorkshire floods), extreme water waves & wave-energy devices.

Overview of wave-energy research

- ▶ Jonathan Bolton provides a worthwhile and excellent overview of wave-energy research in his introductory chapter, including on the contraction device.
- ▶ Bolton (2026, PhD thesis, SoM-UoL): *“On the modelling and optimisation of a novel wave-energy device”*.

<https://eprints.whiterose.ac.uk/>

- (i) incoming wave (hydro)dynamics,
- (ii) (vertical) buoy motion, and
- (iii) power generation.



Buoy in contraction device (3 buoy shapes, 6 frequencies, 8+ resistances):

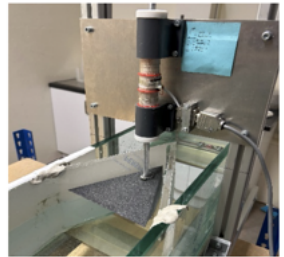


Figure 3.1: (a) – Waveflap driven by servo motor (red); (b) – Power supply and control box, green button generates waves, switch allows manual waveflap adjustments; (c) – Buoy, V-channel and induction coils (red) setup.

Two devices: contraction buoy & elastomer sheets

3 Elastomer sheets; Eco-Flex Sail, Eco-Flex Surface, PVC Surface:

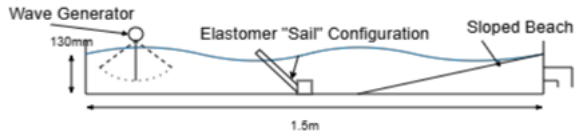


Figure 5.1 - Schematic of the elastomer energy harvesting experimental setup including sloped beach absorber position.

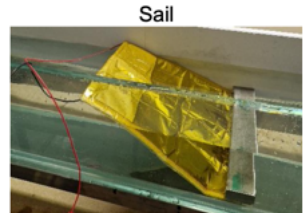
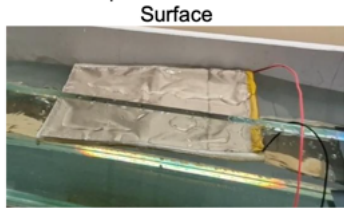
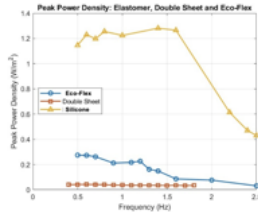
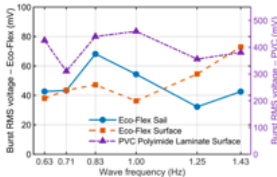
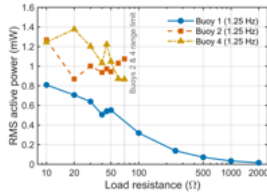
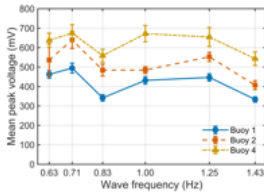


Figure 5.2 - Surface and Sail elastomer geometry configurations used in this chapter.

Preliminary outcomes

Different wave amplitudes; plots need power input normalisation
(standing/progressive waves):

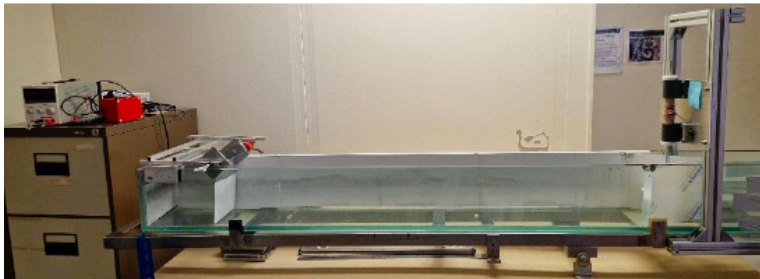


Froude scaling

Equate Froude numbers $Fr = \frac{v_{full}}{\sqrt{gL_{full}}} = \frac{v_{scale}}{\sqrt{gL_{scale}}}$ with

$s_f = L_{scale}/L_{full} < 1$ (ratio length scales), then:

- ▶ Velocity $v_{scale}/v_{full} = \sqrt{L_{scale}/L_{full}} = \sqrt{s_f} < 1$.
- ▶ Period or time with $v = L/T$ gives $T_{scale}/T_{full} = \sqrt{s_f} < 1$.
- ▶ Force $F = ma$ with $a = v/T$ gives: $F_{scale}/F_{full} = s_f^3 < 1$
- ▶ Power $P = Fv$ gives: $P_{scale}/P_{full} = s_f^{3.5} < 1$.



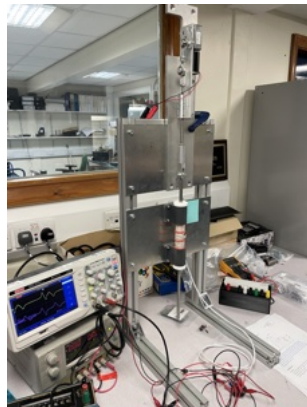
Froude scaling applied: from scale to full model

- ▶ Scale model $L_{mod} = 0.2\text{m}$, full device $L_{full} = 8\text{m}$.
- ▶ $s_f = 0.025, 1/s_f = 40$.
- ▶ Scale buoy power: 1.2mW ; estimate full buoy power: $\sim 500\text{W}$.
- ▶ Yield in passive set-up is very low: **PTO-controller needed; MOSFET, add “negative resistance”?**
- ▶ Scale PVC power: $\sim 1.3\text{W}/\text{m}^2$ (no load); estimate full PVC sheet power: $\sim 31.5\text{kW}$.
- ▶ Cor-Power Ocean buoy 0.3MW . **Factor 10 off. Optimized impedance-matching circuit needed?**

Outline: partitioning & main advances

We have made the following advances for the buoy-generator sub-model (ii-iii):

- ▶ Experimental validation of the model hanging from a driven moving point in its **dry setting**.
- ▶ Same length induction coils placed in parallel and rectified generate most power; three-phase rectifier is best.
- ▶ **NB. Furthered by shown recent advances** in ME-projects.



Grand variational principle of wave-to-wire model

Equations of motion follow from variational principle (**red**=waves, **blue**=buoy, **green**=EM-generator, coupling, B. et al. 2019):

$$\begin{aligned}
 0 = & \rho_0 \delta \int_0^T \int_0^{L_x} \int_{R(t)}^{l_y(x)} \int_0^h -(\partial_t \phi + \frac{1}{2} |\nabla \phi|^2) dz - gh(\frac{1}{2} h - H_0) \\
 & - \frac{1}{2\gamma} \left(F_+ (\gamma(h - h_b) - \lambda)^2 - \lambda^2 \right) dy dx \\
 & \underline{MW\dot{Z} - \frac{1}{2} MW^2 - MgZ + (L_i I - \underline{K(Z)})\dot{Q} - \frac{1}{2} L_i I^2} dt \quad (1)
 \end{aligned}$$

velocity $u = \nabla \phi(x, y, z, t)$, depth $h(x, y, t)$, rest depth H_0 , e.g. buoy $h_b(Z, y) = Z - K_h - \tan \theta (L_y - y)$, piston $R(t)$, coupling function $\gamma_m G(Z) = K'(Z)$, buoy mass M , keel height K_h , buoy coordinate $Z(t)$, buoy velocity $W(t) = \dot{Z}$, charge $Q(t)$, current $I(t) = \dot{Q}$.

Wave-to-wire: PDEs

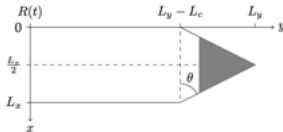
- Potential-flow water-wave dynamics (Laplace equation in interior, kinematic & Bernoulli equations at free surface):

$$\delta\phi : \nabla^2\phi = 0 \quad \text{in} \quad \Omega$$

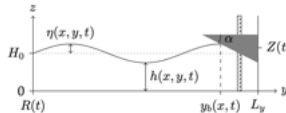
$$(\delta\phi)|_{z=h} : \partial_t h + \nabla\phi \cdot \nabla h = \phi_z \quad \text{at} \quad z = h$$

$$\delta h : \partial_t\phi + \frac{1}{2}|\nabla\phi|^2 + g(z - H_0) - \lambda = 0 \quad \text{at} \quad z = h.$$

- Coupled **elliptic Laplace equation** to **hyperbolic free-surface equations**, plus a (Lagrange) **multiplier** λ .



(b) Top view of the tank and buoy, outlining the tank's dimensions and how the buoy fits the shape of the contraction.



(c) Side view at time t , with the buoy constrained to move vertically.

Wave-to-wire: inequality constraint & ODEs

- Karush-Kuhn-Tucker inequality conditions satisfied at every space-time x, y, t -position are (Burman et al. 2023):

$$\delta\lambda : \lambda = -\max(0, -\gamma(h_b - h) - \lambda) = -\tilde{F}_+(-\gamma(h_b - h) - \lambda) \\ \implies \underline{h_b(Z, y) - h(x, y, t) \geq 0}, \lambda \leq 0, \lambda(h - h_b) = 0.$$

- Add resistance R_i, R_c & Shockley load $V_s(|I|)$ to submodel:

$$\delta W : \dot{Z} = W,$$

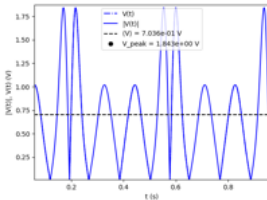
$$\delta Z : M\dot{W} = - \underline{Mg - \gamma_m G(Z)I} - \rho_0 \int_0^{L_x} \int_0^{l_y(x)} \lambda dy dx$$

$$\delta I : \dot{Q} = I,$$

$$\delta Q : L_i \dot{I} = \underline{\gamma_m G(Z) \dot{Z}} - (R_i + R_c)I - \frac{I}{|I|} V_s(|I|).$$

Preliminary experimental validation buoy-generator

- ▶ Modeled/observed outputs $V(t)$ from one-coil (or 3 coils in series) with 1 bridge rectifier & 3 magnets.
- ▶ **Simulation** of one-coil model with ideal bridge-rectifier:
 $1/T_{period} = 2.6\text{Hz}$, $R = 100\Omega$, $l_2 = 0.035\text{m}$. Peak/mean values: (1.843, 0.736)V.
- ▶ **Laboratory observation**: (1.884, 0.921)V for 0.8s.



Experimental validation buoy-generator

Table: Measurements for resistances/loads $R = 50, 100, 900\Omega$ hm.
Set-ups: **S series**: 3 coils in series; **3ph**: three-phase rectified of 3 coils;
1ph: 3 coils put in parallel after rectifying each coil; and,
only using *Top, Middle or Bottom* coil.

Name 1 magnet	Max, Mean V	Name 3 magnets	Max, Mean V	Comment
1-100-1ph.csv	1.222,0.348	3-100-1ph.csv	2.341,1.122	2nd best best glitch 2nd best
1-100-3ph.csv	1.750,0.725	3-100-3ph.csv	2.500, 1.528	
1-100-Bot.csv	nan, nan	3-100-Bot.csv	2.155,1.073	
1-100-Mid.csv	1.046,0.337	3-100-Mid.csv	2.346,0.850	
1-100-S.csv	0.824,0.306	3-100-S.csv	1.884,0.921	
1-100-Top.csv	0.877,0.300	3-100-Top.csv	1.789,0.733	
1-50-1ph.csv	1.107,0.317	3-50-1ph.csv	2.163,1.079	2nd best best
1-50-3ph.csv	1.532,0.482	3-50-3ph.csv	2.500,1.435	
1-50-Bot.csv	0.985,0.258	3-50-Bot.csv	1.896,0.927	
1-50-Mid.csv	0.885,0.299	3-50-Mid.csv	2.045,0.781	
1-50-S.csv	0.635,0.256	3-50-S.csv	1.437,0.703	
1-50-Top.csv	0.767,0.276	3-50-Top.csv	1.584,0.666	

Discretisation strategy: Bouncing Ball –Step-1

- **Step-1:** Define the variations of the augmented Lagrangian $L[Z, W, \lambda]$ for bouncing ball dynamics:

$$0 = \delta \int_0^T L[Z, W, \lambda] dt = \delta \int_0^T W \dot{Z} - \frac{1}{2} W^2 - Z - \frac{1}{2\gamma} \left(F_+(-\gamma Z - \lambda)^2 - \lambda^2 \right) dt.$$

- Definition of variations:

$$\delta \int_0^T L[Z, W, \lambda] dt = \lim_{\epsilon \rightarrow 0} \int_0^T \frac{L[Z + \epsilon \delta Z, W + \epsilon \delta W, \lambda + \epsilon \delta \lambda] - L[Z, W, \lambda]}{\epsilon} dt$$

with perturbation functions $(\delta Z)(t) = \delta Z$, $(\delta W)(t) = \delta W$, $(\delta \lambda)(t) = \delta \lambda$ & end-point conditions $\delta Z(0) = \delta Z(T) = 0$.

- Resulting dynamics:

$$\delta W : \quad \dot{Z} = W$$

$$\delta Z : \quad \dot{W} = -1 + \tilde{F}_+(-\gamma Z - \lambda)$$

$$\delta \lambda : \quad \lambda = -\tilde{F}_+(-\gamma Z - \lambda) = -F_+(-\gamma Z - \lambda)$$

with $\tilde{F}_+(Q) \equiv F_+(Q)F'_+(Q) = F_+(Q) = \max(Q, 0)$.

Discretisation strategy: BB –Step-2

Step-2: In contact dynamics in continuum dynamics, the function $F_+(Q)$ or $\tilde{F}(Q)$ with argument Q often approximated or smoothed, see also ML, by e.g.:

► approximations:

$$\text{Smooth : } \lambda_1(Z) \approx -ae^{-\gamma Z/b} = -\tilde{F}_+(Q = -\gamma Z - \lambda), Q = b \ln(\tilde{F}_+(Q)/a) + \tilde{F}_+(Q) \quad (2)$$

$$\text{Asymp. : } \lambda_3(Z) \approx a \min\left(0, \frac{Z}{b} + \frac{Z|Z|}{d}\right), \tilde{F}_+(Q) = \max\left(0, Q + \frac{(d\Gamma - d\sqrt{\Gamma^2 + 4aQ/d})}{2a/\gamma}\right) \quad (3)$$

$$C_2 : \lambda_4(Z) \approx a \left\{ \frac{Z}{b}, P(Z; \delta), 0 \right\} \quad \text{with } P(Z) \in C_2\text{-spline}, \tilde{F}(Q), F(Q) \text{ implicit.} \quad (4)$$

► Geometrically, these are solutions $\lambda = \lambda(Z)$ to $\lambda \leq 0, Z \geq 0, \lambda Z = 0$, i.e. dilineating the fourth quadrant.

Discretisation strategy: BB –Step-3, new Lagrangian

- **Step-3:** Define the variations of the new Lagrangian $L[Z, W]$ for bouncing ball dynamics:

$$0 = \delta \int_0^T L[Z, W, \lambda] dt = \delta \int_0^T W \dot{Z} - \frac{1}{2} W^2 - Z - \int_0^Z \lambda_i(\tilde{Z}) d\tilde{Z} dt.$$

- Equations:

$$\delta W : \quad \dot{Z} = \frac{\partial H}{\partial W} = W$$

$$\delta Z : \quad \dot{W} = -\frac{\partial H}{\partial Z} = -1 - \lambda_i(Z).$$

- Hamiltonian dynamics for functions $F = F[Z, W]$ with skew-symmetric bracket $\{F, H\}$:

$$\frac{dF}{dt} = \{F, H\} = \frac{\partial F}{\partial Z} \frac{\partial H}{\partial W} - \frac{\partial F}{\partial W} \frac{\partial H}{\partial Z}, \quad (5a)$$

satisfying skew-symmetry $\{F, H\} = -\{H, F\}$ s.t.

$dH/dt = \{H, H\} = 0$ (and Jacobi's identity).

Discretisation strategy: BB –Step-4, AVF-E

- **Step-4:** AVF energy discrete-time conservation uses the skew-symmetry of the bracket alongside linear expansion $Z(s) = Z^n + s(Z^{n+1} - Z^n)$, $W(s) = W^n + s(W^{n+1} - W^n)$ and $t = s\Delta t$, $s \in [0, 1]$ (e.g. Cotter 2023):

$$\begin{aligned}\dot{Z} &= \frac{1}{\Delta t} \frac{dZ(s)}{ds} = \frac{Z^{n+1} - Z^n}{\Delta t} = \int_0^1 \frac{\partial H[Z(s), W(s)]}{\partial W(s)} ds \\ &= \int_0^1 W(s) ds = \frac{1}{2}(W^n + W^{n+1}) \\ \dot{W} &= \frac{1}{\Delta t} \frac{dW(s)}{ds} = \frac{W^{n+1} - W^n}{\Delta t} = \int_0^1 \frac{\partial H[Z(s), W(s)]}{\partial Z(s)} ds = \int_0^1 -1 - \lambda_i(Z(s)) ds.\end{aligned}\tag{6}$$

- For $\lambda_3(Z)$, with $d = b$, $Z_-^n = \min(0, Z^n)$:

$$\frac{(W^{n+1} - W^n)}{\Delta t} = -1 - a \left(\frac{(Z_-^{n+1} + Z_-^n)}{2b} - \frac{((Z_-^{n+1})^2 + Z_-^{n+1}Z_-^n + (Z_-^n)^2)}{3d} \right) \begin{cases} 1 & Z^{n+1} = Z^n \\ \frac{Z_-^{n+1} - Z_-^n}{Z^{n+1} - Z^n} & \text{else} \end{cases}$$

Exact integration important to deal with singularity

Discretisation strategy: BB –Step-4, AVF-E

AVF energy discrete-time conservation uses the skew-symmetry of the bracket alongside linear expansion

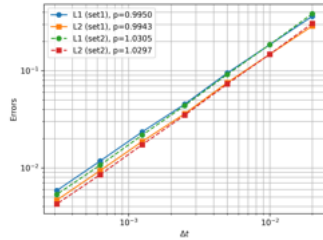
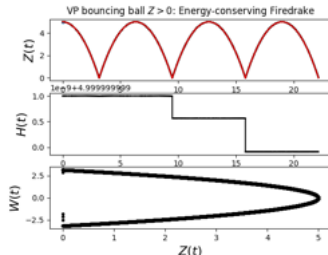
$Z(s) = Z^n + s(Z^{n+1} - Z^n)$, $W(s) = W^n + s(W^{n+1} - W^n)$ and $t = s\Delta t$, $s \in [0, 1]$ (e.g. Cotter 2023). Over one time slab, one proves conservation of energy:

$$\begin{aligned} H[Z^{n+1}, W^{n+1}] - H[Z^n, W^n] &= \int_0^{\Delta t} \frac{dH}{dt} dt = \int_0^1 \frac{dH[Z(s), W(s)]}{ds} ds \\ &= \int_0^1 \frac{\partial H}{\partial Z(s)} \frac{dZ}{ds} + \frac{\partial H}{\partial W(s)} \frac{dW}{ds} ds \\ &= \int_0^1 \frac{\partial H}{\partial Z(s)} (Z^{n+1} - Z^n) + \frac{\partial H}{\partial W(s)} (W^{n+1} - W^n) ds \\ &= \int_0^1 \int_0^1 \frac{\partial H}{\partial Z(s)} \frac{\partial H}{\partial W(\tilde{s})} - \frac{\partial H}{\partial W(s)} \frac{\partial H}{\partial Z(\tilde{s})} ds d\tilde{s} = 0. \end{aligned}$$

Discretisation strategy: –BB Step-5

Step-5, using Firedrake FEM framework solvers:

- Parameters $b = 500, \gamma = 1, \Delta t = 0.035$ ($b\gamma = 500, \gamma = 2, 4$ fine too).
- Relationship between b, γ and Δt .
- **PETSc** solvers (to date): *nest, vinewtonrsls*.



Alternative discretisation –BB

Smooth approximation allows for AVF expression:

$$\tilde{F}_+(Q) = \frac{1}{2}Q + \frac{1}{4}\sqrt{Q^2 + b^2} + \frac{1}{4}\frac{Q^2}{\sqrt{Q^2 + b^2}} \rightarrow_{b \rightarrow 0} \max(0, Q).$$

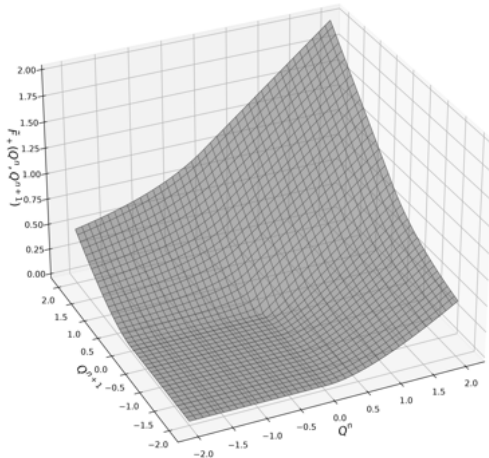
AVF time discretisation reads:

$$Z^{n+1} = Z^n + \frac{1}{2}\Delta t(W^n + W^{n+1}),$$

$$\begin{aligned} W^{n+1} = & W^n - \Delta t + \frac{1}{4}\Delta t(Q^{n+1} + Q^n) \\ & + \frac{1}{4} \left(\frac{((Q^{n+1})^2 + Q^{n+1}Q^n + (Q^n)^2 + b^2 + \sqrt{(Q^{n+1})^2 + b^2})\sqrt{(Q^n)^2 + b^2}}{\sqrt{(Q^{n+1})^2 + b^2} + \sqrt{(Q^n)^2 + b^2}} \right), \quad (7a) \\ 0 = & \frac{1}{2}(\lambda^n + \lambda^{n+1}) + \frac{1}{4}\Delta t(Q^{n+1} + Q^n) \\ & + \frac{1}{4} \left(\frac{((Q^{n+1})^2 + Q^{n+1}Q^n + (Q^n)^2 + b^2 + \sqrt{(Q^{n+1})^2 + b^2})\sqrt{(Q^n)^2 + b^2}}{\sqrt{(Q^{n+1})^2 + b^2} + \sqrt{(Q^n)^2 + b^2}} \right) \end{aligned}$$

with $Q^n = -\gamma Z^n - \lambda^n$ & $Q^{n+1} = -\gamma Z^{n+1} - \lambda^{n+1}$. This is a nonlinear system in $\{Z^{n+1}, W^{n+1}, \lambda^{n+1}\}$.

Alternative discretisation –BB

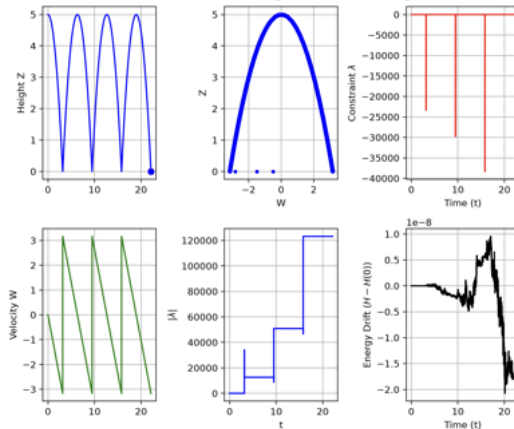


Exact (wire)
and
smoothed
(grey)
 $\tilde{F}_+(Q^n, Q^{n+1})$ –
functions, in
the AVF time
integrator, as
used in the
algorithms
for inequality
constraints.

Alternative discretisation –BB

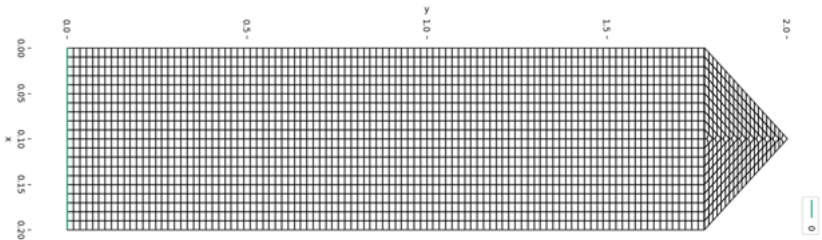
$\sim O(\Delta t)$ when $\gamma = 2.52 \times 10^6 \Delta t^{-1.99}$, $b = 2.83 \times 10^{-3} \Delta t^{0.72}$:

Smoothed AVF Time Integrator – Bouncing Ball



Wave-buoy dynamics: inequality constraints

- ▶ **Inequality constraints** imposed with *derived, smooth penalty superfunction* as Lagrange multiplier (augmented Lagrangian).
- ▶ FEM mesh, e.g., with 1st, 2nd, 3rd or 4th-order polynomials [dofs: (6,22,50,88)k]; AVF-energy-conservative (-) time integration essential; sofar $\lambda_{1,4}(Z)$ work:



Wave-buoy: full dynamics

- ▶ Spectrally accurate Continuous Galerkin (CG) FEM used in space (e.g., 2nd order CG1, CG2, CG3) with time-efficient & error-reducing automation via & implemented in Firedrake environment.
- ▶ **Key take-away**: energy conserving 2nd-order AVF discretisation used for stability, in time $t \in [t^n, t^{n+1} = t^n + \Delta t]$.
- ▶ Used linear-in- s variables (e.g., Cotter 2023)
 $q(s) \equiv (Z(s) = (Z^n + s(Z^{n+1} - Z^n), W(s))$ & likewise for
 $q(s) \equiv \{h(x, y, s), \phi(x, y, s), \psi(x, y, s), Z(s), W(s), I(s)\}$:

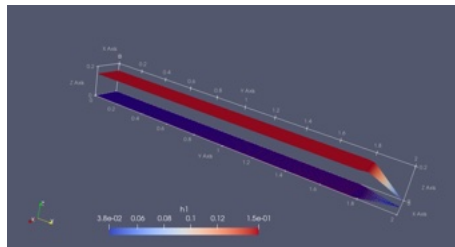
$$\frac{\partial q}{\partial t} = \frac{1}{\Delta t} \frac{dq}{ds} = J(q^{n+1/2}) \int_0^1 \frac{\partial H[q(s)]}{\partial q(s)} ds, \quad (8)$$

with **crucial exact integration**.

- ▶ Polynomials automated; λ_i 's derived/programmed.

Wave-buoy dynamics: initial condition rest state

- ▶ Starting point: used continuation for λ_1 in $b = 10, 100, 1000, 2000, 4000$ then nondimensionalisation. λ_4 best to date. Rest system $\{W, \phi\}$ solved, eqns other variables $\{Z, h, \psi, I\}$ enforced to be linear & zero.
- ▶ Solvers: “bespoke” for rest state, default for dynamics.

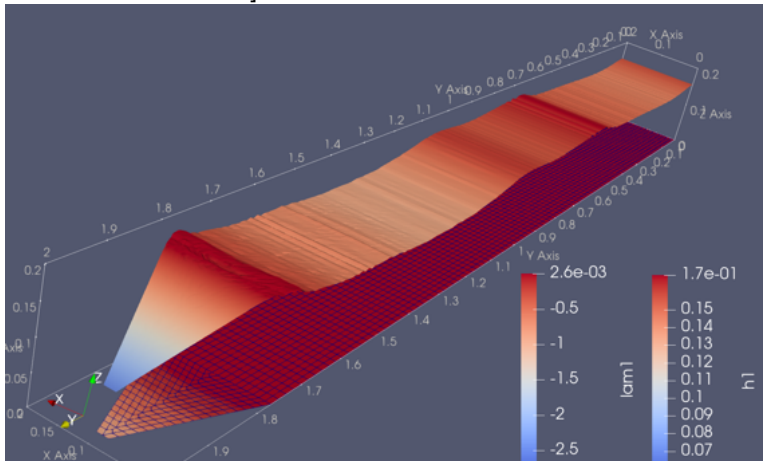


Wave-buoy: full dynamics

- ▶ The conservative dynamics has non-canonical components in the variables $\{W, I\}$, which non-constant $J(Z)$ -bracket terms, involving the nonlinear function $G(Z)$ arising from solving the 3D Maxwell's equations for axisymmetric induction coil(s), are evaluated at $Z^{n+1/2}$ (per review of Cotter 2023).
- ▶ The Variational Boussinesq Model (VBM) is used, a potential flow model with quadratic polynomial accuracy integrated out in z (e.g., Gagarina et al. 2013) with canonical variables $\{h(x, y, t), \phi(x, y, t)\}$ and interior velocity potential degree-of-freedom $\psi(x, y, t)$.

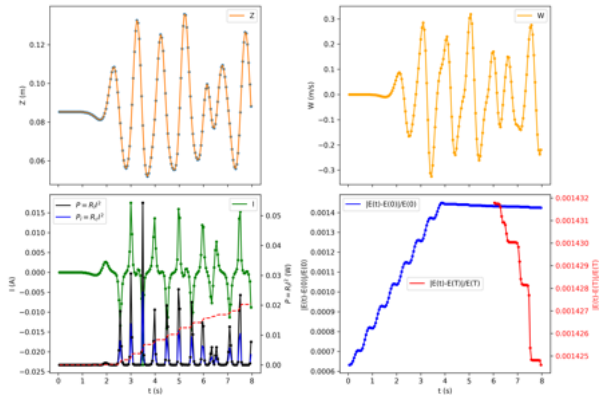
Wave-buoy dynamics: inequality constraints

- Continuous Galerkin CG2 [<https://youtu.be/bBXuj8XBjAk> & <https://youtu.be/9ATf-jcGHs>]:



Wave-buoy dynamics: fully coupled single coil, CG2

- For our device, $\lambda_1(h_b - h)$ works **poorly** since exponential explodes and, hence, $\lambda_4(h_b - h)$ is **best** ($a = g, b = 10\Delta t^{3/2}$). Crucial C_2 smoothness in $\lambda_{1,4}$ needed. λ_4 still noisy:



Thank you very much for your attention ...

- B., Zweers 2013: Proof of principle 2013 https://www.youtube.com/watch?v=SZhe_S0xBWo&t=254s
- B., Kalogirou, Zweers 2019: From bore-soliton-splash to a new wave-to-wire wave-energy model. *Water Waves* 10.1007/s42286-019-00022-9 Bore-soliton-splash: <https://www.youtube.com/watch?v=YSXsXNX4zW0&list=FL6mc7mUa6M4Bo2VkD970urw>
- B., Henry, Kalogirou, Thomas 2020: *Int. Marine Energy J.*
- Choi, Kalogirou, Lu, B., Kelmanson 2024: A study of extreme water waves using a hierarchy of models based on potential-flow theory. *Water Waves* <https://doi.org/10.1007/s42286-024-00084-4>
- B., Bolton, Thompson, Geometric power optimisation of a rogue-wave energy device in a (breakwater) contraction. 8th IEEE Conference on Control Technology & Applications (CCTA) (2024) 6 pp. Preprint <https://eartharxiv.org/repository/view/7260/>
- Lu, Gidel, Choi, B., Kelmanson 2025/26: *Water Waves*. Revision.

